

Table 1: arch sweep: Enzyme/2 — NRMSE (median).

Model	$n=3$	$n=10$	$n=100$	$n=1000$
<i>Full observation</i>				
GRU	—	—	0.044	—
LSTM	—	—	0.065	—
sLSTM	0.533	0.114	0.055	0.009
Transformer	—	—	0.033	—
Mamba	—	—	0.120	—
<i>First-last observation</i>				
GRU	—	—	0.044	—
LSTM	—	—	—	—
sLSTM	—	—	—	—
Transformer	—	—	—	—
Mamba	—	—	—	—

Table 2: arch sweep: Glyc/12 — NRMSE (median).

Model	$n=3$	$n=10$	$n=100$	$n=1000$
<i>Full observation</i>				
GRU	0.610	0.266	0.132	0.065
LSTM	0.371	0.191	0.348	0.073
sLSTM	0.338	0.312	0.128	0.067
Transformer	0.222	0.221	0.106	0.065
Mamba	0.249	0.257	0.154	0.110
<i>First-last observation</i>				
GRU	0.309	0.376	0.564	0.455
LSTM	0.384	0.319	0.525	0.369
sLSTM	0.347	0.793	0.576	0.436
Transformer	0.396	0.359	0.463	0.441
Mamba	0.333	0.455	0.320	0.756

Table 3: arch sweep: Glyc/8 — NRMSE (median).

Model	$n=3$	$n=10$	$n=100$	$n=1000$
<i>Full observation</i>				
GRU	0.146	0.112	0.102	0.090
LSTM	0.269	0.132	0.106	6.4×10^6
sLSTM	0.184	0.111	—	—
Transformer	0.117	0.114	0.110	0.096
Mamba	0.200	0.108	0.104	0.133
<i>First-last observation</i>				
GRU	0.250	0.485	0.679	0.867
LSTM	0.206	0.300	0.675	0.894
sLSTM	0.154	0.704	—	—
Transformer	0.201	0.537	0.551	0.796
Mamba	0.131	0.387	0.512	0.823

Table 4: arch sweep: MOF/4 — NRMSE (median).

Model	$n=3$	$n=10$	$n=100$	$n=1000$
<i>Full observation</i>				
GRU	0.181	0.155	0.107	0.035
LSTM	0.163	0.151	0.115	0.041
sLSTM	0.188	0.155	0.118	0.036
Transformer	0.147	0.126	0.092	0.038
Mamba	0.222	0.152	0.104	0.109
<i>First-last observation</i>				
GRU	1.050	0.540	0.443	0.424
LSTM	0.935	0.681	0.392	0.698
sLSTM	0.848	0.414	0.258	0.262
Transformer	0.831	0.585	0.470	0.571
Mamba	0.798	0.693	0.575	2.0×10^6

Table 5: arch sweep: MOF/6 — NRMSE (median).

Model	$n=3$	$n=10$	$n=100$	$n=1000$
<i>Full observation</i>				
GRU	—	0.067	0.054	0.030
LSTM	—	0.080	0.056	0.029
sLSTM	0.122	0.085	0.054	0.048
Transformer	—	—	0.057	0.034
Mamba	—	—	0.062	0.054
<i>First-last observation</i>				
GRU	—	0.130	0.120	0.092
LSTM	—	—	0.106	0.087
sLSTM	0.351	0.130	0.181	0.115
Transformer	—	—	0.087	0.098
Mamba	—	—	0.114	0.132

Table 6: data ablation: Enzyme/2 — NRMSE (median).

Model	$n=3$	$n=10$	$n=100$	$n=1000$
<i>Full observation</i>				
CMVF	0.526	0.148	0.044	0.012
CMVF-L1	0.526	0.148	0.042	0.014
CMVF-L2	—	—	—	0.013
CMVF-unbounded	0.606	0.080	0.046	0.009
NODE-GRU	0.776	0.103	0.077	0.015
NODE-MLP	0.729	0.116	0.180	0.222
NODE-correction	0.424	0.083	0.171	0.219
Global (θ)	0.466	0.348	0.208	0.253
Initial-condition (θ)	0.500	0.180	0.211	0.244
<i>First-last observation</i>				
CMVF	0.526	0.148	0.044	—
CMVF-L1	0.526	0.148	0.042	—
CMVF-L2	—	—	—	—
CMVF-unbounded	0.606	0.080	0.046	—
NODE-GRU	0.776	0.103	0.077	—
NODE-MLP	0.729	0.116	0.180	—
NODE-correction	0.424	0.083	0.171	—
Global (θ)	0.466	0.348	0.208	0.253
Initial-condition (θ)	0.500	0.180	0.211	0.244

Table 7: data ablation: Enzyme/4 — NRMSE (median).

Model	$n=3$	$n=10$	$n=100$	$n=1000$
<i>Full observation</i>				
CMVF	0.031	0.074	0.004	0.003
CMVF-L1	0.038	0.041	0.009	0.006
CMVF-unbounded	0.046	0.032	0.015	0.003
NODE-GRU	0.733	0.139	0.020	0.008
NODE-MLP	0.509	0.184	0.028	0.005
NODE-correction	0.039	0.031	0.033	1.0×10^6
Global (θ)	0.031	0.030	0.036	0.041
Initial-condition (θ)	0.030	0.028	0.034	1.0×10^6
<i>First-last observation</i>				
CMVF	0.030	0.060	0.004	0.003
CMVF-L1	0.038	0.060	0.008	0.007
CMVF-unbounded	0.037	0.024	0.006	1.0×10^6
NODE-GRU	0.520	0.478	0.253	0.232
NODE-MLP	0.279	0.193	0.237	0.263
NODE-correction	0.067	0.050	0.094	0.063
Global (θ)	0.031	0.031	0.035	0.041
Initial-condition (θ)	0.031	0.028	0.035	1.0×10^6

Table 8: data ablation: Enzyme/6 — NRMSE (median).

Model	$n=3$	$n=10$	$n=100$	$n=1000$
<i>Full observation</i>				
CMVF	0.013	0.007	0.002	0.002
CMVF-L1	0.007	0.010	0.008	0.004
CMVF-unbounded	0.011	0.005	0.004	0.002
NODE-GRU	4.3×10^6	1.9×10^6	7.0×10^5	1.4×10^5
NODE-MLP	7.2×10^6	1.0×10^6	4.0×10^5	5.8×10^4
NODE-correction	3.0×10^6	2.4×10^5	8.7×10^4	1.2×10^4
Global (θ)	0.019	0.014	0.013	0.007
Initial-condition (θ)	0.009	0.002	0.003	0.003
<i>First-last observation</i>				
CMVF	0.012	0.009	0.002	0.002
CMVF-L1	0.010	0.010	0.004	0.005
CMVF-unbounded	0.012	0.005	0.004	0.002
NODE-GRU	5.4×10^7	6.6×10^7	1.6×10^7	1.6×10^7
NODE-MLP	1.4×10^7	5.0×10^7	1.5×10^7	1.6×10^7
NODE-correction	7.7×10^6	2.4×10^7	1.9×10^7	1.4×10^7
Global (θ)	0.020	0.021	0.014	0.007
Initial-condition (θ)	0.009	0.002	0.003	0.003

Table 9: data ablation: Glyc/12 — NRMSE (median).

Model	$n=3$	$n=10$	$n=100$	$n=1000$
<i>Full observation</i>				
CMVF	0.610	0.266	0.132	0.065
CMVF-L1	0.259	0.308	0.148	0.068
CMVF-unbounded	0.215	0.266	0.161	0.077
NODE-GRU	0.282	0.314	0.157	0.123
NODE-MLP	0.252	0.312	0.188	0.116
NODE-correction	0.266	0.266	0.152	0.149
Global (θ)	0.334	0.609	0.425	0.160
Initial-condition (θ)	0.253	0.211	0.151	0.162
<i>First-last observation</i>				
CMVF	0.309	0.376	0.564	0.455
CMVF-L1	0.307	0.323	0.469	7.8×10^6
CMVF-unbounded	0.300	0.644	0.617	0.249
NODE-GRU	1.627	2.464	1.307	5.224
NODE-MLP	2.120	2.667	2.999	73.126
NODE-correction	2.021	2.678	0.987	4.664
Global (θ)	0.333	0.553	0.348	0.356
Initial-condition (θ)	0.353	0.272	0.522	0.654

Table 10: data ablation: Glyc/22 — NRMSE (median).

Model	$n=3$	$n=10$	$n=100$	$n=1000$
<i>Full observation</i>				
CMVF	0.375	0.069	0.020	6.5×10^6
CMVF-L1	0.812	0.069	0.025	0.005
CMVF-unbounded	0.095	0.041	0.015	0.003
NODE-GRU	562.012	663.087	506.099	0.130
NODE-MLP	2.337	3.1×10^5	268.049	0.056
NODE-correction	0.285	0.231	1.5×10^3	—
Global (θ)	0.397	0.343	0.255	0.037
Initial-condition (θ)	0.310	0.112	0.027	0.012
<i>First-last observation</i>				
CMVF	0.638	0.273	0.167	0.206
CMVF-L1	0.634	0.387	0.202	0.236
CMVF-unbounded	0.564	0.423	0.280	0.103
NODE-GRU	6.7×10^6	1.4×10^7	1.4×10^7	3.007
NODE-MLP	7.0×10^6	3.0×10^6	1.4×10^6	35.425
NODE-correction	0.436	1.768	5.4×10^6	—
Global (θ)	0.434	0.388	0.363	—
Initial-condition (θ)	0.809	0.270	0.271	—

Table 11: data ablation: Glyc/4 — NRMSE (median).

Model	$n=3$	$n=10$	$n=100$	$n=1000$
<i>Full observation</i>				
CMVF	0.234	0.226	0.099	0.056
CMVF-L1	0.236	0.253	0.124	0.061
CMVF-unbounded	0.261	0.545	0.209	0.066
NODE-GRU	0.239	0.165	0.102	0.072
NODE-MLP	0.268	0.141	0.118	0.103
NODE-correction	0.295	0.100	0.092	0.098
Global (θ)	0.480	0.876	0.494	0.221
Initial-condition (θ)	0.267	0.225	0.183	0.216
<i>First-last observation</i>				
CMVF	0.493	1.247	1.757	3.994
CMVF-L1	0.347	0.859	0.991	1.299
CMVF-unbounded	2.589	2.058	1.370	26.702
NODE-GRU	0.329	0.438	0.360	0.244
NODE-MLP	0.532	0.572	0.471	6.843
NODE-correction	0.378	1.114	0.999	0.318
Global (θ)	0.565	1.062	0.817	0.758
Initial-condition (θ)	0.523	0.775	0.931	0.791

Table 12: data ablation: Glyc/8 — NRMSE (median).

Model	$n=3$	$n=10$	$n=100$	$n=1000$
<i>Full observation</i>				
CMVF	0.146	0.112	0.102	0.090
CMVF-L1	0.181	0.146	0.138	0.079
CMVF-unbounded	0.192	0.078	0.099	0.062
NODE-GRU	0.169	0.135	0.117	0.102
NODE-MLP	0.271	0.112	0.108	0.093
NODE-correction	0.247	0.169	0.136	0.119
Global (θ)	0.569	1.015	0.712	0.266
Initial-condition (θ)	0.182	0.442	0.175	0.202
<i>First-last observation</i>				
CMVF	0.250	0.485	0.679	0.867
CMVF-L1	0.279	0.772	0.704	1.292
CMVF-unbounded	0.251	0.593	0.639	0.985
NODE-GRU	0.377	0.419	0.769	0.617
NODE-MLP	0.311	0.378	1.812	2.911
NODE-correction	0.902	2.396	1.246	5.578
Global (θ)	0.564	1.013	0.653	1.437
Initial-condition (θ)	0.161	0.461	2.949	3.646

Table 13: data ablation: MOF/12 — NRMSE (median).

Model	$n=3$	$n=10$	$n=100$	$n=1000$
<i>Full observation</i>				
CMVF	0.086	0.014	0.008	0.007
CMVF-L1	0.167	0.027	0.020	0.009
CMVF-unbounded	0.122	0.021	0.017	0.007
NODE-GRU	0.450	0.339	0.133	0.047
NODE-MLP	0.543	0.272	0.130	0.040
NODE-correction	0.200	0.111	0.055	0.015
Global (θ)	0.193	0.160	0.080	0.016
Initial-condition (θ)	0.094	0.025	0.010	0.009
<i>First-last observation</i>				
CMVF	0.119	0.112	0.044	0.011
CMVF-L1	0.119	0.113	0.050	0.019
CMVF-unbounded	0.173	0.080	0.039	0.028
NODE-GRU	3.777	3.715	3.643	3.855
NODE-MLP	1.358	2.121	2.438	1.765
NODE-correction	0.185	0.314	0.185	0.181
Global (θ)	0.169	0.167	0.127	0.051
Initial-condition (θ)	0.142	0.141	0.027	0.024

Table 14: data ablation: MOF/4 — NRMSE (median).

Model	$n=3$	$n=10$	$n=100$	$n=1000$
<i>Full observation</i>				
CMVF	0.181	0.155	0.107	0.035
CMVF-L1	0.210	0.147	0.113	0.045
CMVF-L2	—	—	—	0.038
CMVF-unbounded	0.534	0.133	0.135	0.043
NODE-GRU	0.407	0.159	0.120	0.051
NODE-MLP	1.045	0.132	0.204	0.196
NODE-correction	0.401	0.131	0.151	—
Global (θ)	0.593	0.501	0.313	0.145
Initial-condition (θ)	0.178	0.159	0.115	0.149
<i>First-last observation</i>				
CMVF	1.050	0.540	0.443	0.424
CMVF-L1	1.043	0.602	0.491	0.461
CMVF-L2	—	—	—	—
CMVF-unbounded	0.735	0.424	0.605	0.115
NODE-GRU	1.989	3.658	9.673	3.283
NODE-MLP	0.871	3.890	3.127	1.753
NODE-correction	0.979	0.865	0.487	0.631
Global (θ)	0.948	0.668	0.743	0.887
Initial-condition (θ)	0.743	0.571	0.703	0.840

Table 15: data ablation: MOF/6 — NRMSE (median).

Model	$n=3$	$n=10$	$n=100$	$n=1000$
<i>Full observation</i>				
CMVF	0.156	0.067	0.054	0.030
CMVF-L1	0.250	0.075	0.057	0.045
CMVF-unbounded	0.196	0.079	0.064	0.034
NODE-GRU	0.393	0.181	0.159	0.053
NODE-MLP	0.408	0.182	0.135	0.084
NODE-correction	0.317	0.140	0.069	0.060
Global (θ)	0.240	0.255	0.096	0.095
Initial-condition (θ)	0.111	0.154	0.076	0.094
<i>First-last observation</i>				
CMVF	0.338	0.130	0.120	0.092
CMVF-L1	0.273	0.144	0.076	0.080
CMVF-unbounded	0.264	0.132	0.082	0.094
NODE-GRU	1.540	1.173	2.057	3.051
NODE-MLP	5.559	16.193	11.289	3.485
NODE-correction	0.328	0.309	0.206	0.265
Global (θ)	0.229	0.240	0.116	0.178
Initial-condition (θ)	0.265	0.172	0.184	0.159

Table 16: data ablation: MOF/8 — NRMSE (median).

Model	$n=3$	$n=10$	$n=100$	$n=1000$
<i>Full observation</i>				
CMVF	0.215	0.052	0.019	0.011
CMVF-L1	0.303	0.114	0.026	0.018
CMVF-unbounded	0.200	0.118	0.026	0.015
NODE-GRU	0.370	0.246	0.117	0.054
NODE-MLP	0.388	0.216	0.188	0.067
NODE-correction	0.346	0.173	0.078	0.045
Global (θ)	0.300	0.265	0.164	0.084
Initial-condition (θ)	0.249	0.130	0.072	0.082
<i>First-last observation</i>				
CMVF	0.265	0.151	0.060	0.035
CMVF-L1	0.289	0.156	0.092	0.067
CMVF-unbounded	0.268	0.152	0.111	0.043
NODE-GRU	3.369	3.087	44.252	22.203
NODE-MLP	3.183	10.429	9.217	7.295
NODE-correction	0.409	0.332	0.267	0.658
Global (θ)	0.272	0.259	0.178	0.117
Initial-condition (θ)	0.240	0.160	0.193	0.177